

Day 16 — Sun 20 Sep

Countdown: 6 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🕒 GK — Geography

Source: General/03_GK_CAPSULE.md

3.5 Geography of Nagaland

Feature	Detail
Terrain	Hilly, part of the Naga Hills / Patkai ranges; young fold mountains
Highest peak	Mount Saramati (~3,841 m) △ verify altitude, on the Myanmar border , Kiphire district
Dzükou Valley	Famous valley on the Nagaland–Manipur border near Kohima; the Dzükou lily ; trekking
Japfü peak	Second-highest, near Kohima; famous rhododendron trees
Major rivers	Doyang (longest river <i>in</i> Nagaland △ verify), Dhansiri , Tizu , Zungki , Milak
River drainage	Doyang & Dhansiri drain westward to the Brahmaputra system; Tizu drains eastward to the Chindwin (Myanmar) — a good discriminator fact
Intanki National Park	In Peren district — the state's only national park △ verify sole-status
Fakim Wildlife Sanctuary	Kiphire district
Other protected areas	Puliebadze WLS, Rangapahar WLS △ verify
Climate	Monsoon; heavy rainfall; pleasant summers in the hills

6.2 Rivers

System	Rivers
Himalayan (perennial)	Ganga, Yamuna, Brahmaputra, Indus — snow-fed, perennial
Peninsular (seasonal)	Godavari ("Dakshin Ganga", longest peninsular river), Krishna, Kaveri, Narmada, Tapi (west-flowing into Arabian Sea)
North-East	Brahmaputra (Assam); Nagaland's rivers feed both the Brahmaputra and the Chindwin (see §3.5)

6.3 Climate & soils

- **Climate:** tropical monsoon; the **South-West Monsoon (June–Sept)** delivers most rainfall; retreating (NE) monsoon Oct–Dec brings rain to the SE coast.

- **Major soils:** **Alluvial** (plains, most fertile, most widespread), **Black/Regur** (Deccan, cotton), **Red** (peninsula), **Laterite** (heavy rainfall areas), **Mountain/Forest** (hills, incl. the North-East).

6.4 Resources (headline)

- India is a major producer of **coal** (Jharkhand, Odisha, Chhattisgarh), **iron ore**, **tea** (Assam, West Bengal), and a leading producer of **milk**, **pulses**, **jute**.
- The NE is rich in **petroleum & natural gas** (Assam — Digboi, Asia's oldest refinery), forests and biodiversity.

6.5 Nagaland geography

Cross-linked to §3.5 — Saramati, Dzükou, Doyang/Dhansiri/Tizu, Intanki NP, the Naga Hills, jhum. Know these both as "Nagaland GK" and as "geography."

Geography — likely exam questions

- [5] Name the major soil types of India with one crop each.
- [5] What is the South-West Monsoon and why is it important?
- [10] Describe the major physical divisions of India.

Geography — MCQ traps (with answers)

Answers

1. Longest peninsular river? → **Godavari** 2. Two west-flowing peninsular rivers? → **Narmada and Tapi**
 3. Highest peak in India? → **Kanchenjunga** (Sikkim) 4. Most widespread soil in India? → **Alluvial** 5.
 Asia's oldest oil refinery? → **Digboi (Assam)** 6. Which monsoon brings most of India's rain? → **South-West Monsoon**

SECTION 7 — GENERAL SCIENCE

Intuition first for the whole section. State-GK science is class-8-to-10 level: definitions, units, one formula, and everyday applications. Do not over-study; nail the standard facts.

☉ Core — IP addressing & subnetting

Source: *_Shared/CORE_05_ComputerNetworks.md*

9. IP addressing — with worked subnetting

This is the highest-scoring numerical topic in the paper. Work every example by hand.

Concept

An **IPv4 address** is a **32-bit logical address** identifying an interface on a network. It is written in **dotted decimal** as four **octets** (0–255): **192.168.10.25**. Every address splits into two parts:

_____ NETWORK ID _____ HOST ID _____

The **network ID** says *which network*; the **host ID** says *which machine on it*. Routers

care only about the network part — which is exactly why the split exists: it keeps routing tables small.

How does a router know where the split is? From the **subnet mask** — a 32-bit number with **1s over the network part and 0s over the host part**, contiguously. Written as `255.255.255.0` or as **slash/CIDR notation** `/24` (the count of 1 bits).

The AND operation is the whole mechanism: to find the network address of a host, do a **bitwise AND of the IP address with the subnet mask**. Say this in every subnetting answer.

Address classes (classful addressing)

Class	Leading bits	First octet range	Default mask	Network/Host split	Networks	Hosts per network	Purpose
A	0	1 – 126	255.0.0.0 (/8)	N.H.H.H	$2^7 - 2 = 126$	$2^{24} - 2 = 16,777,214$	Very large organisations
B	10	128 – 191	255.255.0.0 (/16)	N.N.H.H	$2^{14} = 16,384$	$2^{16} - 2 = 65,534$	Medium/large
C	110	192 – 223	255.255.255.0 (/24)	N.N.N.H	$2^{21} = 2,097,152$	$2^8 - 2 = 254$	Small networks
D	1110	224 – 239	—	—	—	—	Multicast (no host portion)
E	1111	240 – 255	—	—	—	—	Experimental / reserved

Why 127 is missing from Class A: `127.0.0.0/8` is reserved for **loopback** (`127.0.0.1` = localhost). And why "-2" in the host counts: **the all-zeros host address is the network address** and **the all-ones host address is the broadcast address**; neither can be assigned to a machine.

Special and reserved addresses

Address	Meaning
<code>0.0.0.0</code>	"This host" / unspecified; also the default route in a routing table
<code>127.0.0.1</code> (<code>127.0.0.0/8</code>)	Loopback — traffic never leaves the machine
<code>255.255.255.255</code>	Limited broadcast — this network only, never forwarded by routers
Network ID (host bits all 0)	Identifies the network itself; not assignable
Broadcast (host bits all 1)	Directed broadcast to all hosts on that network; not assignable
<code>169.254.0.0/16</code>	APIPA / link-local — self-assigned when DHCP fails. Seeing this means "no DHCP server"
<code>224.0.0.0/4</code>	Multicast

Private address ranges — memorise exactly (RFC 1918)

Class	Range	CIDR	Number of addresses
A	10.0.0.0 – 10.255.255.255	10.0.0.0/8	16,777,216

B	172.16.0.0 – 172.31.255.255	172.16.0.0/12	1,048,576
C	192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536

The Class B range is the one people get wrong: it is **172.16 through 172.31**, *not* 172.16 through 172.255 and *not* only 172.16. Private addresses are **not routable on the Internet**; **NAT** (Network Address Translation) on the edge router translates them to one or a few public addresses, which is why the world has not yet run out of IPv4. **PAT / NAT overload** adds port numbers so that thousands of internal hosts can share **one** public address.

Subnetting

Why subnet? A single Class B network with 65,534 hosts is unusable: one broadcast reaches all 65,534 machines, security is impossible, and troubleshooting is hopeless. **Subnetting borrows bits from the host portion and gives them to the network portion**, creating multiple smaller networks inside one address block. Benefits: reduced broadcast traffic, better security/isolation, easier management, efficient address use.

The four formulas — write them at the top of your answer sheet:

Quantity	Formula
Number of subnets	2^n , where n = number of borrowed (subnet) bits
Number of usable hosts per subnet	$2^h - 2$, where h = number of remaining host bits
Block size (subnet increment)	256 – (value of the interesting octet in the mask) , or equivalently 2^h within that octet
Valid host range	(network + 1) to (broadcast – 1)

The "**interesting octet**" is the last octet of the mask that is neither 255 nor 0.

Mask cheat sheet — learn this cold; it makes every question a lookup:

CIDR	Mask	Block size	Hosts/subnet (usable)
/24	255.255.255. 0	256	254
/25	255.255.255. 128	128	126
/26	255.255.255. 192	64	62
/27	255.255.255. 224	32	30
/28	255.255.255. 240	16	14
/29	255.255.255. 248	8	6
/30	255.255.255. 252	4	2 (used for router-to-router links)
/31	255.255.255. 254	2	0 (special, point-to-point RFC 3021)
/32	255.255.255. 255	1	single host

Octet values in order: **128, 192, 224, 240, 248, 252, 254, 255**. These correspond to 1,2,3,4,5,6,7,8 bits borrowed within an octet.

Worked subnetting example 1 — Class C, 4 subnets

Problem: Divide **192.168.1.0/24** into **4 subnets**. Give the mask, the subnet addresses,

the broadcast addresses and the valid host ranges.

Step 1 — bits to borrow. We need 4 subnets, and $2^n \geq 4 \rightarrow n = 2$.

Step 2 — new mask. Default /24, borrow 2 $\rightarrow$ /26. In binary the fourth octet becomes 11000000 = 192, so the mask is **255.255.255.192**.

Step 3 — hosts per subnet. Remaining host bits $h = 32 - 26 = 6$. Usable hosts = $2^6 - 2 = 62$.

Step 4 — block size. $256 - 192 = 64$. So subnets increment by 64 in the last octet.

Step 5 — tabulate.

Subnet	Network address	First host	Last host	Broadcast
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

Check: 4 subnets $\times$ 62 usable hosts = 248 usable, plus 8 addresses lost to network/broadcast pairs = 256. ✓

Worked subnetting example 2 — Class C, "at least 5 subnets"

Problem: An organisation has 192.168.10.0/24 and needs **at least 5 subnets**. Design it.

Step 1. $2^n \geq 5 \rightarrow n = 2$ gives 4 (not enough), **$n = 3$ gives 8 ✓**

Step 2. Mask = /24 + 3 = /27 $\rightarrow$ fourth octet 11100000 = 224 $\rightarrow$ **255.255.255.224**

Step 3. $h = 32 - 27 = 5 \rightarrow$ hosts = $2^5 - 2 = 30$ **per subnet**

Step 4. Block size = $256 - 224 = 32$

#	Network	Host range	Broadcast
1	192.168.10.0	.1 – .30	.31
2	192.168.10.32	.33 – .62	.63
3	192.168.10.64	.65 – .94	.95
4	192.168.10.96	.97 – .126	.127
5	192.168.10.128	.129 – .158	.159
6	192.168.10.160	.161 – .190	.191
7	192.168.10.192	.193 – .222	.223
8	192.168.10.224	.225 – .254	.255

Five subnets are used; **three remain spare** for future growth. Note the trade-off you should comment on: 8 subnets of 30 hosts wastes fewer addresses than 4 subnets of 62 if the departments are small, but caps each subnet at 30 machines.

Worked subnetting example 3 — find the network from a host address

Problem: A host has IP **192.168.20.130** with mask **255.255.255.192**. Find (a) the subnet address, (b) the broadcast address, (c) the valid host range, (d) how many hosts per subnet, (e) is 192.168.20.191 a valid host address?

Step 1 — CIDR. 255.255.255.192 → 192 = 11000000 = 2 bits → /26

Step 2 — block size. 256 – 192 = 64. Subnet boundaries in the last octet: 0, 64, 128, 192.

Step 3 — locate 130. 128 ≤ 130 < 192 → the host lives in the 128 block.

(a) Subnet address = 192.168.20.128

(b) Broadcast = next boundary – 1 = 192 – 1 = 192.168.20.191

(c) Valid hosts = 192.168.20.129 to 192.168.20.190

(d) $2^6 - 2 = 62$ hosts

(e) No — .191 is the broadcast address of this subnet, not assignable.

Verification by AND (show this working — it earns marks):

IP	192.168.20.130 = 11000000.10101000.00010100.10000010
Mask	255.255.255.192 = 11111111.11111111.11111111.11000000
AND	_____
Net	192.168.20.128 = 11000000.10101000.00010100.10000000 ✓

Worked subnetting example 4 — Class B subnetting

Problem: Given 172.16.35.75/20, find the subnet address, broadcast address, valid host range, number of subnets and hosts per subnet.

Step 1 — the mask. /20 = 20 ones = 11111111.11111111.11110000.00000000 = 255.255.240.0. The interesting octet is the third.

Step 2 — borrowed bits. 172.16 is Class B (default /16), so $n = 20 - 16 = 4$ borrowed → $2^4 = 16$ subnets.

Step 3 — hosts. $h = 32 - 20 = 12$ → $2^{12} - 2 = 4,094$ hosts per subnet.

Step 4 — block size. 256 – 240 = 16, applied to the third octet. Boundaries: 0, 16, 32, 48, 64, 80, ..., 240.

Step 5 — locate 35 in the third octet. 32 ≤ 35 < 48 → the block is 32.

Answer	Value
Subnet address	172.16.32.0
Broadcast address	172.16.47.255 (next boundary 48, minus one address)
First valid host	172.16.32.1
Last valid host	172.16.47.254
Subnets	16
Hosts per subnet	4,094

The point to explain: when the interesting octet is the third, the fourth octet is *entirely* host bits, so it runs the full 0–255 within each block. That is why the broadcast is x.x.47.255 and not x.x.35.255.

Worked subnetting example 5 — Class A subnetting

Problem: Given 10.20.30.40/13, find the subnet, broadcast, range and host count.

Step 1 — mask. /13 = 11111111.11111000.00000000.00000000 = 255.248.0.0. Interesting octet = the second.

Step 2 — borrowed bits. Class A default /8, so $n = 13 - 8 = 5$ → $2^5 = 32$ subnets.

Step 3 — hosts. $h = 32 - 13 = 19 \rightarrow 2^{19} - 2 = 524,286$ hosts per subnet.

Step 4 — block size. $256 - 248 = 8$ in the second octet. Boundaries: 0, 8, 16, 24, 32, ...

Step 5 — locate 20. $16 \leq 20 < 24 \rightarrow$ block 16.

Answer	Value
Subnet address	10.16.0.0
Broadcast	10.23.255.255
Host range	10.16.0.1 – 10.23.255.254
Subnets	32
Hosts/subnet	524,286

Worked subnetting example 6 — VLSM design problem

Problem: A company is allocated **192.168.1.0/24**. It has four departments needing **60, 28, 12 and 5** hosts. Design an efficient addressing scheme.

The method — VLSM (Variable Length Subnet Masking): allocate to the **largest requirement first**, using the **smallest mask that fits**, and work downwards. Fixed-length subnetting would force every subnet to /26 and run out; VLSM lets each subnet have its own mask.

Step 1 — size each requirement. Need $2^h - 2 \geq$ requirement.

Dept	Hosts needed	h	Usable = $2^h - 2$	Mask	Block size
A	60	6	62 ✓	/26 (255.255.255.192)	64
B	28	5	30 ✓	/27 (255.255.255.224)	32
C	12	4	14 ✓	/28 (255.255.255.240)	16
D	5	3	6 ✓	/29 (255.255.255.248)	8

Step 2 — allocate sequentially from 192.168.1.0.

Dept	Network	Mask	First host	Last host	Broadcast	Usable
A (60)	192.168.1.0/26	255.255.255.192	.1	.62	.63	62
B (28)	192.168.1.64/27	255.255.255.224	.65	.94	.95	30
C (12)	192.168.1.96/28	255.255.255.240	.97	.110	.111	14
D (5)	192.168.1.112/29	255.255.255.248	.113	.118	.119	6
<i>Spare</i>	192.168.1.120 – 192.168.1.255					136 addresses free

Step 3 — router links. WAN point-to-point links need exactly 2 hosts → use /30 (255.255.255.252, block size 4). Carve them from the spare space:

192.168.1.120/30 (hosts .121, .122, broadcast .123),

192.168.1.124/30 (hosts .125, .126, broadcast .127).

Comment for the examiner: classful/FLSM design at /26 would give only 4 equal subnets of 62 hosts — enough by count, but department D would waste 57 addresses and there would be no room for the /30 WAN links. **VLSM's advantage is address-space efficiency.** Its requirement: routing protocols must be **classless** (send the mask with the update) — **RIPv2, OSPF, EIGRP, BGP** support VLSM; **RIPv1 and IGRP do not.**

Worked example 7 — supernetting / CIDR aggregation

Problem: Aggregate `192.168.8.0/24`, `192.168.9.0/24`, `192.168.10.0/24` and `192.168.11.0/24` into one route.

Method: convert the varying octet to binary and find the **common prefix**.

```
8 = 0000 1000
9 = 0000 1001
10 = 0000 1010
11 = 0000 1011
    ^^^^^^
    first 6 bits are common (000010)
```

Common bits in the third octet = 6. Total prefix = 8 + 8 + 6 = **/22**.

Mask = 255.255.**252**.0. **Supernet = 192.168.8.0/22**, covering 192.168.8.0 through 192.168.11.255 — **1,024 addresses, 1,022 usable**.

Why this matters: **CIDR (Classless Inter-Domain Routing, RFC 1519)** abolished the rigid A/B/C boundaries. **Route aggregation / summarisation** lets one routing-table entry replace four, which is the only reason the global BGP table is manageable. Note the requirement: **blocks must be contiguous and the starting address must be a multiple of the block size** (8 is a multiple of 4 ✓).

IPv4 vs IPv6

IPv4's 32 bits give 4.3 billion addresses — exhausted. **IPv6** uses **128 bits**, written as eight groups of four hex digits separated by colons:

`2001:0db8:85a3:0000:0000:8a2e:0370:7334`, shortened by dropping leading zeros in each group and replacing **one** run of all-zero groups with `::` → `2001:db8:85a3::8a2e:370:7334`. The `::` may appear **only once**, otherwise the expansion is ambiguous.

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Address space	$2^{32} \approx 4.3 \times 10^9$	$2^{128} \approx 3.4 \times 10^{38}$
Notation	Dotted decimal , <code>192.168.1.1</code>	Colon hexadecimal , <code>2001:db8::1</code>
Header	Variable, 20–60 bytes , 13 fields	Fixed 40 bytes , 8 fields + extension headers
Header checksum	Present	Removed (Layer 2 and 4 already check) — faster routers
Fragmentation	By sender and routers	By the source only (Path MTU Discovery)
Broadcast	Yes	No broadcast — replaced by multicast and anycast
Address types	Unicast, multicast, broadcast	Unicast, multicast, anycast
Configuration	Manual or DHCP	SLAAC (stateless autoconfiguration) or DHCPv6
Security	IPSec optional	IPSec built in (mandatory in the original spec)
ARP	Uses ARP	Uses NDP (Neighbor Discovery Protocol, ICMPv6)

QoS	ToS field	Flow Label + Traffic Class
Loopback	127.0.0.1	::1
Default route	0.0.0.0/0	::/0
NAT	Essential	Largely unnecessary

Transition mechanisms to name: **dual stack**, **tunnelling** (6to4, Teredo — IPv6 packets encapsulated in IPv4), and **translation** (NAT64).

MAC address, ARP and RARP

A **MAC address** (physical/hardware address) is a **48-bit (6-byte)** address **burned into the NIC** by the manufacturer, written as **12 hex digits**: **00:1A:2B:3C:4D:5E**. The **first 3 bytes (24 bits) are the OUI** — Organizationally Unique Identifier, assigned to the vendor; the last 3 bytes are the vendor's serial number. The broadcast MAC is

FF:FF:FF:FF:FF:FF.

	MAC address	IP address
Layer	2 (Data Link)	3 (Network)
Size	48 bits (6 bytes, hex)	32 bits IPv4 (4 bytes, decimal)
Assigned by	Manufacturer , burned into hardware	Network administrator / DHCP
Changes?	Permanent (though spoofable in software)	Changes when the device moves network
Scope	Local link only — never crosses a router	End to end, globally routable
Also called	Physical/hardware/burned-in address	Logical address

Why both? Here is the sentence that answers the whole question: **the IP address says where the packet is ultimately going; the MAC address says where it goes next**. As a packet crosses the internet, the **source and destination IP addresses never change**, but the **source and destination MAC addresses are rewritten at every hop** — each router strips the old frame and builds a new one for the next link.

Protocol	Full name	Maps	Used by	Note
ARP	Address Resolution Protocol	IP → MAC	A host that knows the IP of a neighbour but needs its MAC to build a frame	Broadcasts "Who has 192.168.1.5? Tell 192.168.1.10"; the owner unicasts the reply. Results cached in the ARP cache (<code>arp -a</code>)
RARP	Reverse ARP	MAC → IP	A diskless workstation that knows only its own MAC and needs an IP at boot	Obsolete — replaced by BOOTP and then DHCP
ICMP	Internet Control Message Protocol	—	Error reporting and diagnostics	<code>ping</code> (echo request/reply), <code>traceroute</code> (TTL exceeded), destination unreachable
DHCP	Dynamic Host Configuration Protocol	Assigns IP + mask + gateway + DNS	Automatic host configuration	Four-step DORA : D iscover (client broadcast) → O ffer (server) → R quest (client) → A cknowledge (server). Ports 67/68

ARP spoofing / poisoning (relevant to your Forensic paper): ARP has **no authentication**, so an attacker can send unsolicited ARP replies claiming the gateway's IP maps to the attacker's MAC. Victims then send all their traffic to the attacker — a **man-in-the-middle** attack, and the basis of most LAN packet sniffing on switched networks.

Likely exam questions

- **[5]** What is an IP address? Explain the classes of IP addresses with their ranges.
- **[5]** What is a subnet mask? Explain the AND operation with an example.
- **[5]** List the private IP address ranges. Why are they needed?
- **[5]** Differentiate between MAC address and IP address.
- **[5]** What is ARP? How does it differ from RARP?
- **[5]** Compare IPv4 and IPv6.
- **[10]** Given the address 192.168.20.130 with mask 255.255.255.192, determine the subnet address, broadcast address, valid host range and number of hosts. Show all working.
- **[10]** Divide 192.168.1.0/24 into 8 subnets. Give the mask, and tabulate every subnet's network address, host range and broadcast address.
- **[15]** Explain IP addressing in detail: classes, subnet masks, subnetting, CIDR and VLSM. An organisation with 192.168.1.0/24 has departments needing 60, 28, 12 and 5 hosts. Design a complete VLSM addressing scheme with full working.
- **[15]** Explain classful and classless addressing. Discuss the exhaustion of IPv4 and the solutions adopted — CIDR, NAT, private addressing and IPv6 — and compare IPv4 with IPv6.

MCQ traps

Trap	Correct answer
"Class B first-octet range?"	128–191 . Class A 1–126 , Class C 192–223 .
"Why is 127 excluded?"	Loopback .
"Hosts in a Class C network?"	254 (not 256).
"255.255.255.240 = ?"	/28 , block size 16, 14 hosts .
"Private Class B range?"	172.16.0.0 – 172.31.255.255 (not 172.16–172.255).
"MAC address size?"	48 bits . IPv4 = 32, IPv6 = 128 .
"ARP maps?"	IP → MAC . RARP is the reverse.
"Number of subnets from 3 borrowed bits?"	2³ = 8 .
"APIPA range?"	169.254.0.0/16 .
"Broadcast of 192.168.1.64/26?"	192.168.1.127 .
"Mask for a 2-host WAN link?"	/30 — 255.255.255.252.
"Does the IP address change at each hop?"	No — the MAC does.
"DHCP sequence?"	DORA .
"IPv6 has broadcast?"	No — multicast and anycast only.